

Guillermo Terrén-Serrano, PhD

University of California Santa Barbara, Santa Barbara, CA, USA

+1 (505) 450-6215
guillermoterren@ucsb.com

Webpage: gterren.github.io
GitHub: github.com/gterren

Education

Ph.D. in Electrical Engineering “pass with distinction”	2016 - 2022
<i>Electrical & Computer Engineering Department, University of New Mexico, NM, USA.</i>	<i>Advisor: Prof. Martínez-Ramón</i>
M.Sc. in Power & Energy	2015 - 2016
<i>Electrical & Computer Engineering Department, University of New Mexico, NM, USA.</i>	<i>Advisor: Prof. Martínez-Ramón</i>
B.Sc. in Electrical Engineering	2006 - 2011
<i>Escuela de Ingeniería y Arquitectura, Universidad de Zaragoza, Zaragoza, Spain.</i>	<i>Advisor: Prof. Ibáñez-Carabantes</i>

Experience

CNSI Climate Innovation Postdoctoral Fellow	2022 - Present
<i>Environmental Markets Lab (emLab) and Environmental Studies, University of California Santa Barbara, CA, USA.</i>	<i>Supervisor: Prof. Deshmukh</i>
Postdoctoral Research Associate	2022 - 2023
<i>Statistics and Applied Probability Department, University of California Santa Barbara, CA, USA.</i>	<i>Supervisor: Prof. Ludkovski</i>
Graduate Research Assistant	2015 - 2022
<i>CEET and Electrical & Computer Engineering Department, University of New Mexico, NM, USA.</i>	
Project Engineer	2012 - 2013
<i>Enerland Group, Zaragoza, Spain.</i>	<i>Supervisor: Mr. Ballet-Sala</i>
Undergraduate Research Assistant	2012
<i>Electronic Engineering Department, Institute of Technology for the Development (LACTEC), PR, Brazil.</i>	<i>Supervisor: Prof. Zambenedetti</i>

Peer-Reviewed Publications

- [Ter+26b] **Cost-Optimal and Robust Decarbonization Pathways for India’s Electricity System under Uncertainty.** *Submitted*, 2026. G. Terrén-Serrano, R. Deshmukh, et al.
- [Ter+26a] **Probabilistic Day-Ahead Forecasting of System-Level Renewable Energy and Electricity Demand.** *Nature Communications*, 2026. G. Terrén-Serrano, R. Deshmukh and M. Martínez-Ramón.
- [TL25] **Extreme Day-Ahead Renewables Scenario Selection in Power Grid Operations.** *Applied Energy*, 2025. G. Terrén-Serrano and M. Ludkovski.
- [Pal+23] **Advances in Solar Forecasting: Computer Vision with Deep Learning.** *Advances in Applied Energy*, 2023, Q. Paletta, G. Terrén-Serrano, et al.
- [TM23a] **Deep Learning for Intra-Hour Solar Forecasting with Fusion of Features Extracted from Infrared Sky Images.** *Information Fusion*, 2023. G. Terrén-Serrano and M. Martínez-Ramón.
- [TM23b] **Detection of Multiple Wind Velocity Fields using Ground-based Infrared Sky Images.** *Knowledge-Based Systems*, 2023. G. Terrén-Serrano and M. Martínez-Ramón.
- [TM23c] **Kernel Learning for Intra-Hour Solar Forecasting with Infrared sky images and Cloud dynamic feature extraction.** *Renewable and Sustainable Energy Reviews*, 2023. G. Terrén-Serrano and M. Martínez-Ramón.
- [TM23d] **Processing of Global Solar Irradiance and Ground-Based Infrared Sky Images for Solar Nowcasting and Intra-Hour Forecasting Applications.** *Solar Energy*, 2023. G. Terrén-Serrano and M. Martínez-Ramón.
- [TM22] **Geospatial Perspective Reprojections for Ground-Based Sky Imaging Systems.** *IEEE Transactions on Geoscience and Remote Sensing*, 2022. G. Terrén-Serrano and M. Martínez-Ramón.
- [TM21a] **Comparative Analysis of Methods for Cloud Segmentation in Ground-Based Infrared Images.** *Renewable Energy*, 2021. G. Terrén-Serrano and M. Martínez-Ramón.
- [TM21b] **Multi-Layer Wind Velocity Field Visualization in Infrared Images of Clouds for Solar Irradiance Forecasting.** *Applied Energy*, 2021. G. Terrén-Serrano and M. Martínez-Ramón.
- [Ter+21] **Girasol, a Sky Imaging and Global Solar Irradiance Dataset.** *Data in Brief*, 2021. G. Terrén-Serrano, A. Bashir, et al.
- [Mam+19] **An Experimental Method to Merge Far-field Images from Multiple Long-Wave Infrared Sensors for Short-term Solar Forecasting.** *Solar Energy*, 2020. A. Mammoli, G. Terrén-Serrano, et al.
- [Gar+18] **Evaluation of Dimensionality Reduction Methods Applied to Numerical Weather Models for Solar Radiation Forecasting.** *Engineering Applications of Artificial Intelligence*, 2018. O. García-Hinde, G. Terrén-Serrano, et al.

Working Papers

Resource Adequacy in Electricity Systems under Demand Growth and Decarbonization. Expected in spring '26. G. Terrén-Serrano, A. Sharma, et al.

Dynamic Intraday Probabilistic Forecasting Update of Renewable Power Generation. Expected in summer '26. G. Terrén-Serrano and M. Ludkovski.

Reliability Contribution of Renewable Energy Resources in Large Grids under Decarbonization. Expected in fall '26. G. Terrén-Serrano et al.

Employment, Health, and Environmental Effects of India's Power System Low-Carbon Transition. Expected in winter '27. W. T. Wang and G. Terrén-Serrano, et al.

Conference Proceedings

Day-Ahead Operational Forecast of Aggregated Solar Generation Assimilating Mesoscale Meteorology Information. *IEEE PES Grid Edge Technologies Conference & Exposition*, San Diego, USA, 2025. G. Terrén-Serrano, R. Deshmukh, and M. Martínez-Ramón.

Explicit Basis Function Kernel Methods for Cloud Segmentation in Infrared Images. *Energy Reports*, 2022. G. Terrén-Serrano and M. Martínez-Ramón.

Wind Flow Estimation in Thermal Sky Images for Sun Occlusion Prediction. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022. G. Terrén-Serrano and M. Martínez-Ramón.

Segmentation Algorithms for Ground-Based Infrared Cloud Images. *IEEE PES ISGT-Europe*, Espoo, Finland, 2022. G. Terrén-Serrano and M. Martínez-Ramón.

Selected Presentations

2025 Forecasting & Markets Workshop (organized by ESIG). *Presentation*. Nashville, TN, USA.

2025 UCSB Institute for Energy Efficiency (IEE). *Seminar*. CA, USA.

2024 ARPA-E Energy Innovation Summit. *Exhibition booth*. Dallas, TX, USA.

2024 Macro-Energy Systems (MES) Workshop. *Flash talk + poster*. Princeton University, NJ, USA.

2024 Massachusetts Institute of Technology (MIT) Energy Initiative. *Seminar*. MA, USA.

2023 International Energy Workshop (IEW). *Presentation*. Colorado School of Mines, CO, USA.

Teaching

ES105 Renewable Energy Systems. *Lecturer*. Environmental Studies Program, University of California Santa Barbara.

ECE 203 Circuitry Analysis I. *Teaching Assistant*. Electrical and Computer Engineering (ECE) Department, University of New Mexico (UNM).

ECE 213 Circuitry Analysis II. *Teaching Assistant*. ECE, UNM.

ENG Introduction to Electronic Circuits. *Teaching Assistant*. ECE, UNM.

ECE 314 Signals and Systems. *Teaching Assistant*. ECE, UNM.

ECE 331 Introduction to Data Structures and Algorithms. *Teaching Assistant*. ECE, UNM.

Selected Grants

UCSB CNSI Climate Innovation Found 2023-2024. Funding (\$30K) to accelerate innovations that promote climate resilience; supports computing resources and conference travel.

UCSB IEE Research Seed Program 2022-2023. One-year award (\$50K) to develop software for smart grid control systems, energy storage hierarchies, and market mechanisms; includes work on joint day-ahead solar forecasting.

Selected Honors and Awards

CNSI Climate Innovation Postdoctoral Fellowship 2023-2025. Competitive fellowship supporting research on technologies for California's climate goals; research focuses on extending joint day-ahead forecasting from solar generation (IEE seed program) to wind and electricity demand.

UNM-Avangrid Graduate Scholarship, 2015. Award for graduate study in Electrical and Computer Engineering at the UNM under the King Felipe VI of Spain Endowed Chair, sponsored by the Iberdrola Foundation (Avangrid).

UZ-Santander Bank Americampus Scholarship, 2011. Competitive grant supporting an academic exchange in Latin America focused on sustainable buildings and renewable energy applications.

Institutional Service and Community Involvement

Mentor. David Lopez and Steven De Jesus. UCSB EUREKA! Program 2022 for STEM undergraduate students, Santa Barbara, CA.

Mentor. David Lopez. UCSB CAMP Research School Year Internship 2022, Santa Barbara, CA.

Student Athlete. 2006-2011 Rugby team, Universidad de Zaragoza, Zaragoza, Spain.

Reviewer. Applied Energy, Renewable Energy, Knowledge-Based Systems, Solar Energy, and Data in Brief.

Translator. Parent-teacher conferences for Spanish speakers at South Valley Academy (SVA), Albuquerque, NM.

Skills

OS: Linux, Ubuntu, macOS, Windows, Raspbian.

Programming Languages: Python, R, Matlab, C/C++, LabVIEW, Simulink, HTML, Pascal, Bash.

Scientific Computing (HPC): CPU, GPU, and MPI parallel computing.

Prototyping: Arduino and Raspberry Pi, and 3D printing (CAD).

Libraries: TensorFlow, Keras, PyTorch, OpenCV, Scikit-Learn.

Version Control: Git, Overleaf.

Languages: Spanish (native), Portuguese (fluent).

Professional References

Dr. Ranjit Deshmukh (rdeshmukh@ucsb.edu). Associate Professor. *Bren School of Environmental Science & Management, UCSB, CA, USA.*

Dr. Manel Martínez-Ramón (manel@unm.edu). Professor. *Department of Electrical and Computer Engineering, UNM, NM, USA.*

Dr. Michael Ludkovski (mludkov@ucsb.edu). Professor. *Department of Statistics and Applied Probability, UCSB, CA, USA.*

Dr. Andrea Mammoli (aamammo@sandia.gov). Principal Member. *Sandia National Laboratories, NM, USA.*